

**Министерство науки и высшего образования Российской Федерации
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ ОБРАЗОВАТЕЛЬНОЕ
УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО
ITMO University**

Лабораторная работа №1

По дисциплине Администрирование компьютерных сетей

Обучающиеся Рыбинская Злата Владиславовна, Кудина Вероника
Евгеньевна, Сокович Степан Витальевич, Зеленков Евгений Алексеевич,
Алексеев Николай Сергеевич

Факультет Факультет инфокоммуникационных технологий

Группа K3344

Направление подготовки 45.03.04 Интеллектуальные системы в
гуманитарной сфере

Образовательная программа Интеллектуальные системы в гуманитарной
сфере

Обучающиеся Рыбинская З.В., Сокович С.В., Зеленков Е.А.,
Кудина В.Е. Алексеев Н.С.

Преподаватель

Самохин Н.Ю.

Заполним докер файл для постгреса

```
FROM postgres:15

RUN apt-get update -y && \
    apt-get install -y netcat-openbsd python3-pip curl python3-
psycpg2 python3-venv iputils-ping

RUN python3 -m venv /opt/patroni-venv && \
    /opt/patroni-venv/bin/pip install --upgrade pip && \
    /opt/patroni-venv/bin/pip install patroni[consul] psycpg2-
binary

COPY postgres0.yml /postgres0.yml
COPY postgres1.yml /postgres1.yml

ENV PATH="/opt/patroni-venv/bin:$PATH"

USER postgres
```

Вместо zookeeper было принято решение использовать consul, прежде всего в целях эксперимента

Поэтому для мастера и реплики прокидываем следующие настройки в файлы конфигурации

мастер:

```
scope: my_cluster
name: postgresql0
namespace: /service/
restapi:
  listen: 0.0.0.0:8008
  connect_address: pg-master:8008

consul:
  host: consul:8500
  port: 8500
```

```
bootstrap:
  dcs:
    ttl: 30
    loop_wait: 10
    retry_timeout: 10
    maximum_lag_on_failover: 10485760
    postgresql:
      use_pg_rewind: true
      use_slots: true
      parameters:
        wal_level: replica
        hot_standby: "on"
        wal_keep_segments: 8
        max_wal_senders: 10
        max_replication_slots: 10
        wal_log_hints: "on"
      pg_hba:
        - local all all trust
        - host all all 127.0.0.1/32 trust
        - host replication replicator 0.0.0.0/0 md5
        - host all all 0.0.0.0/0 md5

postgresql:
  listen: 0.0.0.0:5432
  connect_address: pg-master:5432
  data_dir: /var/lib/postgresql/data/postgresql0
  bin_dir: /usr/lib/postgresql/15/bin
  pgpass: /tmp/pgpass0
  authentication:
    replication:
      username: replicator
      password: rep-pass
```

```
  superuser:
    username: postgres
    password: postgres
  parameters:
    unix_socket_directories: '.'

watchdog:
  mode: automatic

tags:
  nofailover: false
  noloadbalance: false
  clonefrom: false
  nosync: false
```

Реплика:

```
scope: my_cluster
name: postgresql1
namespace: /service/
restapi:
  listen: pg-slave:8008
  connect_address: pg-slave:8008

consul:
  host: consul:8500
  port: 8500

bootstrap:
  dcs:
    ttl: 30
    loop_wait: 10
    retry_timeout: 10
    maximum_lag_on_failover: 10485760
```

```
postgresql:
  use_pg_rewind: true
  use_slots: true
  parameters:
    wal_level: replica
    hot_standby: "on"
    wal_keep_segments: 8
    max_wal_senders: 10
    max_replication_slots: 10
    wal_log_hints: "on"
  pg_hba:
    - local all all trust
    - host all all 127.0.0.1/32 trust
    - host replication replicator 0.0.0.0/0 md5
    - host all all 0.0.0.0/0 md5
```

```
postgresql:
  listen: 0.0.0.0:5432
  connect_address: pg-slave:5432
  data_dir: /var/lib/postgresql/data/postgresql1
  bin_dir: /usr/lib/postgresql/15/bin
  pgpass: /tmp/pgpass1
  authentication:
    replication:
      username: replicator
      password: rep-pass
    superuser:
      username: postgres
      password: postgres
  parameters:
    unix_socket_directories: '.'
```

```
watchdog:
```

```
mode: automatic

tags:
  nofailover: false
  noloadbalance: false
  clonefrom: false
  nosync: false
```

Настроим поднятие всех наших сервисов через docker-compose. Вместо haproxy используем nginx.

```
services:
```

```
consul:
  image: consul:1.15
  container_name: consul
  hostname: consul
  ports:
    - "8500:8500"
    - "8600:8600"
  command: "agent -dev -client=0.0.0.0"
  volumes:
    - consul-data:/consul/data

pg-master:
  build: .
  image: localhost/postgres:patroni-consul
  container_name: pg-master
  restart: always
  hostname: pg-master
  environment:
```

```
    POSTGRES_USER: postgres
    POSTGRES_PASSWORD: postgres
    PGDATA: '/var/lib/postgresql/data/pgdata'
  expose:
    - 8008 # Patroni API
  ports:
    - 5433:5432 # Master
  volumes:
    - pg-master:/var/lib/postgresql/data
  command: patroni /postgres0.yml
  depends_on:
    - consul

pg-slave:
  build: .
  image: localhost/postgres:patroni-consul
  container_name: pg-slave
  restart: always
  hostname: pg-slave
  expose:
    - 8008 # Patroni API
  ports:
    - 5434:5432 # Slave
  volumes:
    - pg-slave:/var/lib/postgresql/data
  environment:
    POSTGRES_USER: postgres
    POSTGRES_PASSWORD: postgres
    PGDATA: '/var/lib/postgresql/data/pgdata'
  command: patroni /postgres1.yml
  depends_on:
    - consul
```

```

nginx:
  image: nginx:1.25
  container_name: nginx_balancer
  restart: always
  hostname: nginx
  ports:
    - 5432:5432 # Entry point
    - 8080:8080 # Nginx status page
  depends_on:
    - pg-master
    - pg-slave
  volumes:
    - ./nginx.conf:/etc/nginx/nginx.conf
    - ./nginx-status.conf:/etc/nginx/conf.d/status.conf
  command: [nginx, '-g', 'daemon off;']

volumes:
  consul-data:
  pg-master:
  pg-slave:

```

Сбилдим и поднимем всю нашу инфра

```

C:\Users\alex\uchshiy\University\computerNet>docker-compose up --build
[*] Building 15/15
✓nginx Pulled 18.4s
✓09f376ebb190 Pull complete 14.9s
✓a11fc495bafd Pull complete 15.5s
✓933cc8470577 Pull complete 15.5s
✓999643392f97 Pull complete 15.5s
✓971bb74f4fb12 Pull complete 15.5s
✓45337c09cd57 Pull complete 15.6s
✓de3b062c0af7 Pull complete 15.6s
✓consul Pulled 11.8s
✓d078792c4f91 Pull complete 2.8s
✓fbdcc5b403c6 Pull complete 2.8s
✓c0e228c45cba Pull complete 3.9s
✓d9a4cd1fc71 Pull complete 9.0s
✓a23041e1d950 Pull complete 9.0s
✓27d6d74a7c1d Pull complete 9.0s

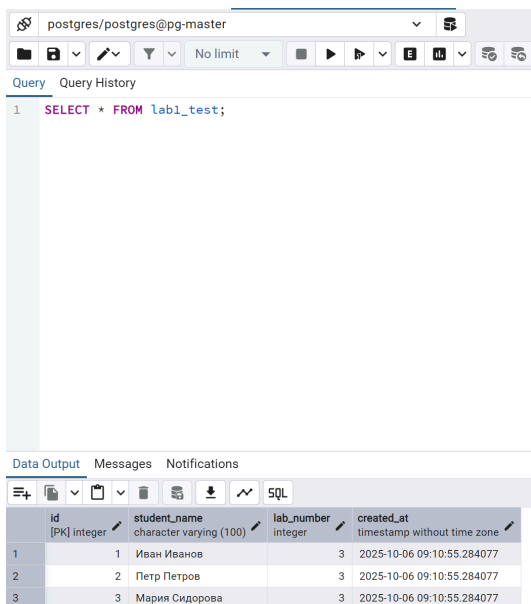
```

Базово проверим по логам, что мастер и слейв живы.

```
PS C:\Users\alexluchshiy\University\computerNet> docker logs pg-master --tail 10
2025-10-06 08:27:37,474 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:27:47,461 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:27:57,455 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:28:07,451 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:28:17,456 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:28:27,455 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:28:37,454 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:28:47,459 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:28:57,457 INFO: no action. I am (postgresql0), the leader with the lock
2025-10-06 08:29:07,456 INFO: no action. I am (postgresql0), the leader with the lock
```

```
PS C:\Users\alexluchshiy\University\computerNet> docker logs pg-slave --tail 10
2025-10-06 08:28:17,962 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:28:27,681 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:28:37,962 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:28:47,836 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:28:57,858 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:29:07,668 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:29:18,139 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:29:28,059 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:29:37,753 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-10-06 08:29:47,645 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
```

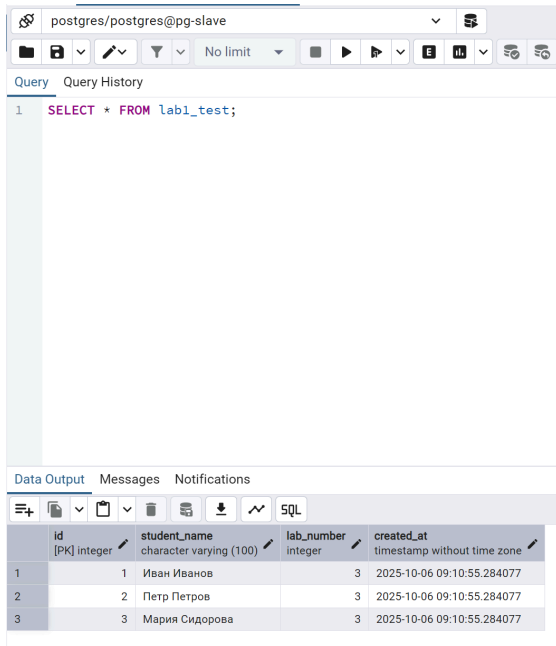
Проверим синхронизацию мастера и реплик путём вставки в мастер и чтения с реплики (коннект напрямую, не через балансер). Создадим отношение lab-test и сделаем вставку 3-х кортежей на мастере.



The screenshot shows a PostgreSQL client interface with a query editor and a results table. The query is `SELECT * FROM lab1_test;`. The results table has four columns: `id` (integer, primary key), `student_name` (character varying (100)), `lab_number` (integer), and `created_at` (timestamp without time zone). The table contains three rows of data.

	id [PK] integer	student_name character varying (100)	lab_number integer	created_at timestamp without time zone
1	1	Иван Иванов	3	2025-10-06 09:10:55.284077
2	2	Петр Петров	3	2025-10-06 09:10:55.284077
3	3	Мария Сидорова	3	2025-10-06 09:10:55.284077

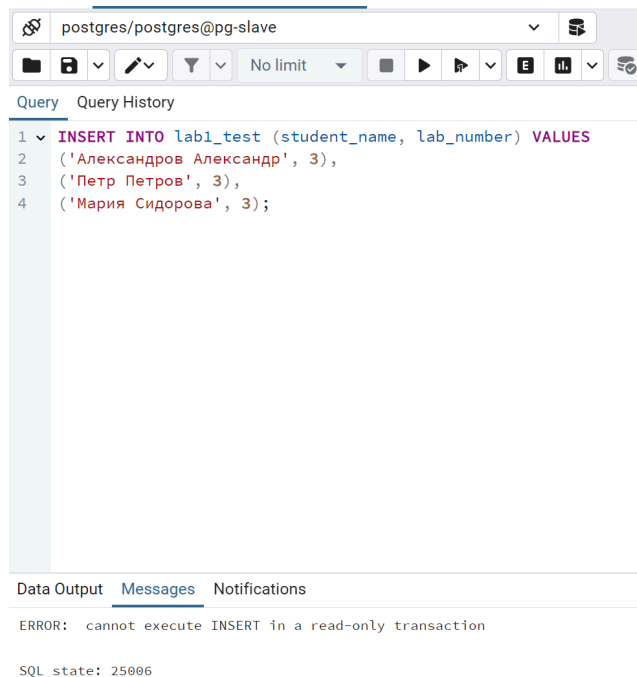
Посмотрим, что имеем на слейве



The screenshot shows a PostgreSQL client window with the connection 'postgres/postgres@pg-slave'. The query history shows a single query: `SELECT * FROM lab1_test;`. The results are displayed in a table with the following data:

	id	student_name	lab_number	created_at
1	1	Иван Иванов	3	2025-10-06 09:10:55.284077
2	2	Петр Петров	3	2025-10-06 09:10:55.284077
3	3	Мария Сидорова	3	2025-10-06 09:10:55.284077

Отлично, синхронизация работает. Также проверим, что реплика является read-only и отсекает все транзакции с обновлением или вставкой соответствующей ошибкой



The screenshot shows the same PostgreSQL client window. The query history shows an `INSERT INTO` statement. The 'Messages' tab is active, displaying an error message: `ERROR: cannot execute INSERT in a read-only transaction`. The SQL state is `25006`.

	id	student_name	lab_number	created_at
1	1	Иван Иванов	3	2025-10-06 09:10:55.284077
2	2	Петр Петров	3	2025-10-06 09:10:55.284077
3	3	Мария Сидорова	3	2025-10-06 09:10:55.284077

ERROR: cannot execute INSERT in a read-only transaction

SQL state: 25006

Конфиг балансировщика выглядит следующим образом:

```
nginx.conf
File Edit View

events {
    worker_connections 1024;
}

stream {
    upstream postgres_backend {
        server pg-master:5432;
        server pg-slave:5432;
    }

    server {
        listen 5432;
        proxy_pass postgres_backend;
        proxy_connect_timeout 1s;
        proxy_timeout 3s;
    }
}
```

Сходим в бд через балансир, создадим еще одну табличку и проверим работу вставки.

```
C:\Users\alexluchshiy\University\computerNet>docker run --rm --network=computerNet_default -e PGPASSWORD=postgres postgres:15 psql -h nginx -U postgres -c "CREATE TABLE balance_test (id SERIAL, data TEXT); INSERT INTO balance_test (data) VALUES ('via nginx'); SELECT * FROM balance_test;"
CREATE TABLE
INSERT 0 1
 id | data
-----+-----
  1 | via nginx
(1 row)
```

В какой-то момент в силу непредвиденных обстоятельств был произведен перезапуск сервисов, поэтому поменялись названия узлов...

Выключаем мастера, проверяем состояние по логам

```
C:\Users\alexluchshiy\University\computerNet>docker stop pg-slave
pg-slave
```

```
PS C:\Users\alexluchshiy\University\computerNet> docker logs pg-slave --tail 10
2025-10-06 13:05:31.778 UTC [21] LOG:  received fast shutdown request
2025-10-06 13:05:31.783 UTC [21] LOG:  aborting any active transactions
2025-10-06 13:05:31.783 UTC [63] FATAL:  terminating connection due to administrator command
2025-10-06 13:05:31.783 UTC [30] FATAL:  terminating connection due to administrator command
2025-10-06 13:05:31.783 UTC [60] FATAL:  terminating connection due to administrator command
2025-10-06 13:05:31.786 UTC [21] LOG:  background worker "logical replication launcher" (PID 38) exited with exit code 1
2025-10-06 13:05:31.789 UTC [23] LOG:  shutting down
2025-10-06 13:05:31.814 UTC [23] LOG:  checkpoint starting: shutdown immediate
2025-10-06 13:05:31.833 UTC [23] LOG:  checkpoint complete: wrote 0 buffers (0.0%); 0 WAL file(s) added, 0 removed, 2 recycled; write=0.005 s, sync=0.001 s, total=0.020 s; sync files=0, lo
ngest=0.000 s, average=0.000 s; distance=16233 kB, estimate=16233 kB
2025-10-06 13:05:31.836 UTC [21] LOG:  database system is shut down
```

Проверим, что реплика переняла на себя нагрузку и стала мастером.

```
PS C:\Users\alexluchshiy\University\computerNet> docker logs pg-master --tail 10
2025-10-06 13:06:33,816 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:06:43,815 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:06:53,816 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:07:03,815 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:07:13,818 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:07:23,814 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:07:33,814 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:07:43,818 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:07:53,817 INFO: no action. I am (postgresl0), the leader with the lock
2025-10-06 13:08:03,810 INFO: no action. I am (postgresl0), the leader with the lock
```

Проверим отказоустойчивость системы путём того, что попросим вставить еще одну строку в таблицку balance_test. Проверка прошла успешно

```
C:\Users\alexluchshiy\University\computerNet>docker run --rm --network=computerNet_default -e PGPASSWORD=postgres postgres:15 psql -h nginx -U postgres -c "INSERT INTO
balance_test (data) VALUES ('test without 1 node'); SELECT * FROM balance_test;"
INSERT 0 1
 id |      data
-----+-----
  1 | via nginx
 34 | test without 1 node
(2 rows)
```